

Temporal Relevance in Semantic Retrieval: Investigating Implicit Temporal Bias and Its Impact on Retrieval-Augmented Generation

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1 Motivation

Modern retrieval-augmented generation (RAG) systems assemble answers by fetching documents from large web indices and conditioning a language model on the retrieved context. Much recent work has focused on improving semantic relevance, whether a document addresses the query's topic. However, the equally important dimension of temporal relevance has received comparatively little systematic attention. While the publication date of a document is an observable and machine-readable attribute, the degree to which semantic retrieval systems implicitly leverage or disregard temporal signals remains largely uncharacterized. This disparity constitutes a meaningful gap in the information retrieval (IR) literature, particularly as RAG systems increasingly rely on retrieved documents to supply factual context to large language models (LLMs).

A central observation motivating this work is that different types of information change at fundamentally different rates. Stable, evergreen knowledge such as our understanding of the biochemical mechanism of photosynthesis remains factually invariant across years or even decades. Time-sensitive information, by contrast, including the composition of governments, current legislation, market prices, or ongoing scientific controversies, may become outdated within months or weeks. Conflating these two regimes is not a neutral choice: forcing a recency bias onto a time-stable query can surface newer but inferior documents, while ignoring recency for a time-sensitive query can produce factually outdated answers. The problem is therefore not merely about fetching recent content, it is about matching retrieval behavior to the temporal profile of the query. The seminar paper will investigate whether semantic retrieval systems exhibit such adaptive behavior implicitly, quantifies the impact of explicit and implicit temporal query signals, and evaluates whether simple recency-based reranking can improve factual correctness in RAG pipelines for time-sensitive queries.

2 Research Questions

RQ1: Implicit Temporal Bias in Semantic Retrieval

Do semantic retrieval systems exhibit implicit temporal retrieval behavior, systematically favoring more recent documents for time-sensitive queries compared to time-stable queries, in the absence of any explicit temporal cues?

Gap and justification: Prior work has characterized recency bias in traditional keyword-based IR systems, yet the extent to which semantic retrieval models internalize temporal sensitivity has not been systematically studied. Because this question can be operationalized as a comparison of the temporal distribution of retrieved documents across query categories, it is well within the scope of a controlled seminar experiment.

RQ2: Effect of Temporal Query Markers on Retrieval

To what extent do implicit and explicit temporal markers in queries influence the retrieval of temporally relevant documents for time-sensitive queries in semantic retrieval systems?

Gap and justification: The literature on query reformulation documents the surface-level effect of adding temporal qualifiers to queries; however, their quantitative impact on the temporal distribution of retrieved documents in dense retrieval settings has not been established. This question can be answered by extending the RQ1 query set with temporally annotated variants and applying the same evaluation metric, keeping all other experimental variables constant.

RQ3: Temporal Reranking and Factual Correctness in RAG

How does temporal reranking based on document recency affect factual correctness in RAG systems for time-sensitive queries?

Gap and justification: RAG systems have been widely evaluated on tasks such as open-domain question answering, however, the specific contribution of recency-based reranking to factual accuracy for time-sensitive queries remains understudied. Answering this question is feasible by combining the retrieval pipeline with an LLM-as-a-judge evaluation protocol against a manually curated ground truth, a methodology well-established in recent RAG evaluation literature.

3 Methodology

All relevant experiments, models and results will be stored in a GitHub¹ repository.

3.1 Dataset

The study will use the stanford-oval/ccnews² corpus, a large-scale collection comprising several million news articles from globally distributed news agencies, distributed as WARC (Web ARChive) files and spanning the years 2016 through 2024. The temporal depth and topical breadth of this corpus make it well-suited for analyzing retrieval behavior across both stable and dynamic information domains. To maintain computational tractability, the dataset will be filtered to a representative subset sampled uniformly across a predefined set of evergreen topic domains (e.g., natural sciences) and time-sensitive topic domains (e.g., political developments), ensuring balanced representation across the corpus's temporal range.

3.2 Query Construction

A controlled query set will be manually developed across three experimental stages. For RQ1, two query categories will be constructed: (i) time-stable queries on topics for which document

¹ <https://git.fim.uni-passau.de/motte/data-science-seminar>

² <https://huggingface.co/datasets/stanford-oval/ccnews>

recency is largely immaterial to factual accuracy, and (ii) time-sensitive queries on inherently dynamic topics, both categories deliberately excluding any temporal markers so that the retrieval system's implicit behavior is isolated. For RQ2, the time-sensitive query set will be systematically extended into three sub-categories: queries without temporal markers, queries with implicit temporal markers (e.g., “current”, “latest”, “recent”), and queries with explicit temporal markers (e.g., “since 2023”, “as of 2024”). For RQ3, a subset of time-sensitive queries will be augmented with manually verified ground-truth answers, enabling factual evaluation of generated responses.

3.3 Evaluation

For RQ1, the temporal distribution of retrieved documents will be quantified using a recency-weighted rank metric applied separately to time-stable and time-sensitive query categories. A statistically significant difference between distributions would constitute evidence of implicit temporal bias. For RQ2, the same metric will be applied across all three temporal marker sub-categories, enabling incremental measurement of the effect of temporal cues. For RQ3, a selected set of time-sensitive queries will be routed through a RAG pipeline in two conditions: without recency-based reranking and with recency-based reranking applied prior to context assembly. The factual correctness of generated answers will be evaluated against ground-truth annotations using either manual assessment or an LLM-as-a-judge protocol, a paradigm that has demonstrated strong correlation with human judgments in recent benchmarking studies.

3.4 Tools and Frameworks

The retrieval pipeline will be implemented using dense retrieval with sentence-transformer embeddings and FAISS-based approximate nearest-neighbor search. Data processing of the CCNews WARC corpus will be performed using Python-based pipelines. An LLM API will be used for both the RAG generation step and the LLM-as-a-judge evaluation component.

4 Timeline

Milestone 1: Data and query foundation: This milestone includes the setup of the CCNews pipeline: a representative subset across your chosen evergreen and time-sensitive topic domains will be filtered. The full query set of time-stable and time-sensitive queries with and without markers for RQ1 and RQ2 will be manually constructed until 30.05.26.

Milestone 2: First retrieval experiments (RQ1 & RQ2): a dense retrieval pipeline will be built and proof-of-concept experiments across all query types will be run until 07.06.26.

Milestone 3: RAG pipeline & ground-truth construction (RQ3): A recency-based reranking component will be integrated into the retrieval pipeline. Time-sensitive queries from the RQ3 subset will be matched to a ground truth for later evaluation until 21.06.26.

5 Limitations

Several limitations constrain the scope and generalizability of this study. The CCNews corpus, while large and multi-lingual, constitutes a curated sample of news media rather than a representation of the open web; findings may therefore not transfer directly to general-purpose web search systems, which exhibit markedly different document distributions and coverage patterns.

Furthermore, the categorization of topic domains into time-sensitive and time-stable is inherently not binary: temporal sensitivity exists on a continuous spectrum and the assignment of topics to either category involves an element of subjective judgment, which may introduce classification noise. The study deliberately confines itself to the analysis of implicit temporal behavior within standard semantic retrieval systems. Temporal retrieval architectures and purpose-built temporal RAG variants, though natural candidates for future work, fall outside the defined scope of the present seminar paper and are not evaluated.

6 Disclaimer

Generative AI was used as assistance during the creation process of this document.